

Context-Dependent Selection Among Established Urban Routing Policies: Where the Decision Value Is, and Where It Is Not

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Abstract

Navigation systems implement established routing objectives — shortest distance, dynamic travel time, capacity-aware load balancing, reliability-aware routing — but commit to one by design rather than by traffic condition. We characterise that choice empirically. In 12,960 SUMO runs over 648 controlled contexts spanning demand, signal timing, guidance penetration, information lag, disruption and alternative capacity, with every comparison paired under common random numbers, we separate two questions that are routinely conflated: whether the preferred policy varies with context, and whether exploiting that variation is worth anything.

It does vary. Three of four policies are strictly best in some context by a seed-resolved margin, the winner map needs more than one factor, and 73.0% of contexts have a non-singleton Pareto set over journey time, CO₂ and stopped delay — with the reliability-aware policy minimising stopped delay in 66% of contexts while almost never minimising journey time. Exploiting it is worth 0.45 s per vehicle (0.12%), because the best fixed policy is near-dominant, with an upside 18× its downside; choosing the wrong fixed policy costs 46.4%. Within that margin, a selector trained to predict the winner’s identity is *more accurate and more expensive* than the fixed policy (67% vs 65% accuracy; 0.77 s vs 0.45 s regret), whereas one trained to predict each policy’s *advantage* is less accurate and closes 55.8% of the available gap. Adaptation helps in-distribution and harms under every extrapolation and shift tested; a support-and-confidence gate prevented all of that harm by abstaining, correctly, because the conformal interval is wider than the effect.

The operational finding is that the decision worth making here is which fixed policy to run and which criterion to optimise, not which policy to run in which context — and that this can be established before any model is built, by comparing the best fixed policy with the retrospective best and both with the noise floor of the same comparison. We introduce no routing, learning or multi-criteria decision algorithm.

Keywords: routing-policy selection; traffic regimes; decision regret; algorithm selection; microsimulation; distribution shift; selective prediction.

1 Introduction

Choosing a route and choosing a *routing policy* are different decisions. A navigation service may minimise distance, or estimated travel time, or may spread demand across alternatives, or may avoid corridors whose travel time is unpredictable. Each of these is an established objective with its own literature. Production systems commit to one and apply it uniformly.

Whether that commitment costs anything depends on the traffic, not on the planner. A distance-minimising policy is unimprovable when the shortest corridor is uncongested and indefensible when it is saturated. A policy that reacts to measured travel times is useful when congestion builds slowly and counterproductive when a large guided cohort acts together on information that is already stale [1, 2]. Whether these regimes exist, where their boundaries lie, and whether they can be recognised before a policy is activated, are empirical questions.

This paper answers them for a controlled microsimulation environment. The question is:

Given a traffic context observable before a routing policy is activated, which established policy should be activated, why is it the right one in that regime, and when should the selector decline to adapt at all?

Research questions. **RQ1** Under which traffic, signal, information, penetration, capacity and disruption conditions do established routing policies exhibit complementary strengths? **RQ2** Can the preference boundaries between them be explained by mechanistic traffic variables rather than only by a statistical learner? **RQ3** Can a machine-learning selector identify useful selection structure beyond a strong mechanistic baseline? **RQ4** When the operating context lies outside the training support, or uncertainty is too high, can the system abstain safely and fall back to a fixed policy?

Contributions. (1) A systematic characterisation of the operating regions in which four established routing policies exhibit complementary performance, under congestion, signal timing, information lag, guidance penetration, disruption and alternative capacity. (2) A mechanism-informed explanation of policy switching using interpretable dimensionless traffic descriptors, so that the boundary is a statement about traffic and not about a model. (3) A decision-oriented context-aware selector, evaluated by decision regret against both a strong fixed baseline and the mechanistic rule — the latter being the comparison that matters. (4) A selective mechanism that recognises support and uncertainty limits and abstains to a fixed policy, evaluated separately under five kinds of distribution shift.

We introduce no routing algorithm, no learning algorithm and no multi-criteria decision method. Every policy and every model class used here is established and cited. The contribution is the characterisation, the mechanism, the decision evaluation and the safety envelope.

2 Related work

Routing objectives. Distance- and travel-time-minimising route choice sits within the equilibrium tradition founded by Wardrop [3]. Load-balancing route guidance, in which vehicles are distributed over a set of travel-time-acceptable alternatives according to predicted link load, is established; we use the flow-balanced k -shortest-path construction of Pan et al. [4]. Reliability-aware guidance, in which a dispersion term is added to expected travel time, follows Sen et al. [5]. Eco-routing, which minimises an emissions proxy, is established [6]; whether it is *distinct* from time-based routing in a given network is an empirical question that we test rather than assume.

Information, penetration and instability. That providing information to drivers need not reduce congestion has been understood since Arnott et al. [1], and has been given a sharp game-theoretic form by Acemoglu et al. [2]. The benefit of guidance depends on how much of the demand is guided [7]. These results motivate treating penetration and information lag as *factors*, not as fixed implementation details.

Per-instance algorithm selection. Choosing among algorithms by instance features is Rice’s problem [8], with a mature methodology [9, 10, 11] and standard reference points — the virtual best and single best solver — from the benchmarking literature [12]. Instance-space analysis asks where in a feature space each algorithm is strong [13], which is precisely the question RQ1 asks of routing policies. Evaluating a predictor by the cost of the decision it induces rather than by its prediction error is the predict-then-optimize distinction of Elmachoub and Grigas [14].

Abstention and distribution shift. Deciding not to decide is classical [15]. Split conformal prediction supplies distribution-free intervals [16, 17], but its coverage guarantee rests on exchangeability, which covariate shift breaks [18, 19]. We therefore use a conformal gate as a calibrated in-distribution instrument and a separate support gate for extrapolation, and say so explicitly rather than claiming that conformal prediction solves the out-of-distribution problem.

Positioning. We do not claim that regime-dependent routing has never been studied. Each component above has precedent. What is offered is the combination: an explicitly established policy portfolio selected per context in a physical microsimulation, evaluated by decision regret against both a best fixed policy *and* a mechanistic rule, with mechanism-informed features, distribution shift reported separately by shift type, and a support-and- confidence gate that abstains.

3 Problem formulation

A *context* s is a set of operating conditions known before a policy is activated. A *policy* $a \in \mathcal{P}$ maps individual vehicles to routes by its own objective. The decision is the map $\pi : s \mapsto a$, taken once per context.

Two levels. At level 1 a context selects a policy; at level 2 the selected policy routes vehicles. This is not route prediction, not vehicle-level action learning, not signal control and not joint routing–signal optimisation. The policies are fixed, published objectives; only the choice among them is learned.

Decision cost and regret. Writing $C(s, a)$ for the cost of running policy a in context s ,

$$R(s, a) = C(s, a) - \min_{b \in \mathcal{P}} C(s, b). \quad (1)$$

Two reference points from the algorithm-selection literature [12] bound what is achievable. The **single best policy** (SBS) is the one fixed policy with the lowest mean cost, chosen on training contexts only; it is what a deployed fixed-policy system achieves, and it is the baseline any selector must beat to be worth deploying. The **virtual best policy** (VBS) chooses the best policy for each context using that context’s realised outcomes. *VBS is retrospective. It is not an algorithm, it cannot be deployed, and it is used here only as an upper reference.* The distance between them is the headroom available to any selector.

Criteria. Three criteria are measured (Table 3): mean journey time C1, CO₂ emissions C2, and total stopped delay C3 — the time vehicles spend stationary in queues. Policies are first compared by Pareto dominance over all three. Scalarisation is used only for sensitivity, through three declared normalised weight profiles. No monetary value of time, value of reliability or carbon price is used: no verified source for them was available in this study, and inventing one would make the conclusion an artefact of the invented number.

4 Experimental methodology

Network. Figure 1 shows the environment. One origin–destination pair is served by three corridors between a common diverge and a common merge: a short, signalised, two-lane central arterial; a longer and slower signalised street with one or two lanes; and a longest, uninterrupted single-lane bypass. Four minor streets carry independent demand through the same signals, so that a routing policy that loads a corridor also delays traffic that is not being routed. Because every junction permits straight-through movements only, the set of loopless paths from origin to destination is exactly these three: the candidate path catalogue is complete, not a k -shortest approximation.

Two structural choices are deliberate. The destination portal gives each approach a dedicated exit lane, so that an alternative corridor is not penalised by junction priority rather than by its own capacity; and the diverge is not subject to a turn-radius speed ceiling, which in schematic node geometry would cap the bypass at a value belonging to the drawing rather than to the road. Both were identified by capacity validation before the experiment was frozen.

Measured capacity. Each corridor was driven to saturation alone and its discharge measured. Queue-discharge saturation flow is 1761 veh/h/lane on the arterial and 1596 veh/h/lane on the slower street, consistent to within 1.8% across green ratios; the uninterrupted bypass carries 1500 veh/h, below queue-discharge saturation flow as free-flow headways require. Capacity is therefore a measured property of this network, not a textbook constant asserted about it, and the demand levels of Table 2 follow from it.

Policies. The four policies are given in Table 1 and Figure 2. All read traffic state from one shared observation buffer: a policy acting at time t may read link travel times sampled no later than $t - \Delta$, over a window of 120 s, and falls back to free-flow values before any admissible sample exists. No policy sees the future and none sees any realised outcome of the run it is part of. Policy parameters were frozen before the experiment: admissibility tolerance $\varepsilon = 0.20$ and a 120 s assignment window for P3, $\lambda = 1.0$ for P4.

Each policy was validated before use. Twelve known-answer tests check that the observation lag hides newer samples, that the free-flow fallback engages, that P1 ignores congestion, that P2 follows the observed mean, that P4 prefers the steadier of two paths with equal means, and that P3 respects its admissibility tolerance.

A two-path integration test inside the simulator, with equal path lengths and a 2:1 capacity ratio, confirms that P3 splits demand 1.97 against the capacity ratio of 2.00 while P1 commits the entire cohort to one path.

Factors and common random numbers. Six factors (Table 2, Figure 3) are crossed fully, giving 648 contexts, each run with 5 seeds and all four policies. The vehicle set, departure times, guided/unguided labelling, habitual corridor choice, background traffic and the incident realisation are functions of the seed and the context alone, never of the policy. Two runs of a context under different policies therefore face identical vehicles, and every comparison in this paper is paired.

Unguided drivers are not re-routed by anything: they follow a habitual corridor drawn once from a frozen logit over free-flow path times. This is what makes penetration meaningful — at penetration p , a fraction $1 - p$ of corridor demand is inert with respect to the policy under test.

Disruption as risk, not as a known event. When the incident regime is active, one lane closure is drawn — location, onset and duration — from a declared distribution seeded by the run seed. Within a context and seed the realisation is identical across all four policies, preserving the pairing exactly; across the seeds of a context it differs, so the context carries genuine travel-time risk that no policy can know in advance. An incident that occurred at an identical time and place in every replication would be a known event rather than a risk, and could not test a reliability-aware policy at all.

Statistical treatment. The experimental unit is the context, not the vehicle. An advantage of a over b is *resolved* only when, across the paired seed differences, $|\bar{d}| > 2 \text{SE}(\bar{d})$. Unresolved differences are treated as ties throughout, so that no claim in this paper rests on a difference the seeds cannot separate.

5 Mechanistic regime model

A selector that fires on raw demand explains nothing. We therefore express each context through dimensionless descriptors with physical meaning, all computable before policy activation: network saturation $x_1 = D / \sum_c \text{cap}_c$; saturation of the shortest corridor $x_2 = D / \text{cap}_C$; alternative capacity share x_3 ; effective green ratio x_4 ; penetration x_5 ; information staleness $x_6 = \Delta / T_C^{\text{ff}}$, the lag as a fraction of the free-flow trip time it is meant to describe; incident regime x_7 ; and controlled-flow saturation $x_8 = p D / \text{cap}_C$, the share of the shortest corridor’s capacity that the policy actually commands.

Corridor capacity uses the measured saturation flows, effective green and lane count, so x_1 , x_2 and x_8 are genuine degrees of saturation rather than scaled demand. A route-overlap index is not used: the three corridors share only their access and egress edges, so overlap is structurally constant here and carries no information.

The mechanistic rule B1 is an axis-aligned region rule on x_1 – x_8 , of depth at most two, fitted on training contexts only. Its thresholds are reported as *brackets between adjacent sampled levels*, never as exact continuous values: the experiment samples a grid, and a grid cannot locate a boundary more finely than its own spacing.

6 Context-aware selector

Target. The selector predicts *advantage*, not cost. For each policy a ,

$$\Delta_a(x) = C_{\text{SBS}}(x) - C_a(x), \quad (2)$$

where the single best policy is determined on *training contexts only*. A positive Δ_a means a is better in that context than the fixed policy a deployed system would otherwise run, so the quantity being learned is exactly the quantity a decision-maker needs. Selection is $\arg \max_a \hat{\Delta}_a(x)$.

Model. Gradient-boosted decision trees [20, 21], one regressor per policy, with hyperparameters fixed before evaluation and never tuned on a test fold. The data are a few hundred context-level instances with full feedback from every policy; a high-capacity model is neither needed nor defensible at this sample size. We do not use graph neural networks, deep networks, reinforcement learning or contextual bandits: the decision is a one-shot choice over four alternatives with complete feedback, which none of those formalisms is required for.

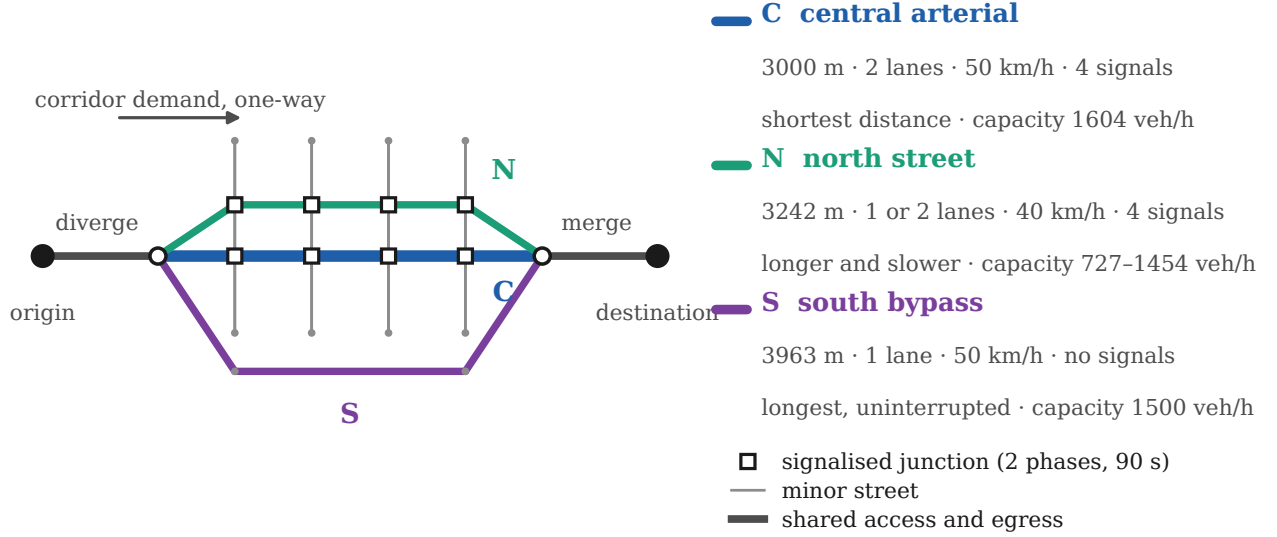


Figure 1: The scenario network. One origin–destination pair is served by three structurally different corridors between a common diverge and merge, crossed by four minor streets carrying independent demand through the same signals. Geometry, lane counts and speed limits are read from the built network file; capacities are measured, not assumed.

Baseline ladder. B0 is the best fixed policy; B1 the mechanistic rule; B2 a depth-3 decision tree; B3 multinomial logistic regression; B4 the GBDT advantage selector; B5 the selective variant below. VBS is reported as a reference. **The primary comparison is B4 against B1**, not B4 against B0: B0 is a low bar, and beating it demonstrates only that the decision varies with context at all. If the mechanistic rule wins, the finding is that the decision structure is simple enough that a learner adds nothing — and that is reported as the result.

Leakage discipline. Evaluation is leave-one-context-out. Inside each fold, standardisation, the identity of the SBS, tree thresholds, conformal quantiles and the support envelope are all recomputed from training data alone. No seed of a held-out context appears in training. No realised outcome of a test context is used as a feature; the eight descriptors are functions of the context definition and cannot be computed from an outcome.

Abstention. Two gates must both pass before the selector acts. The *support* gate measures the mean distance to the five nearest training contexts in standardised descriptor space and abstains above the 95th percentile of the training fold’s own such distances. The *confidence* gate is a split-conformal interval on the predicted advantage ($\alpha = 0.10$, calibrated on 30% of the training fold); the selector acts only when the lower bound on the advantage over the fixed policy is positive. If either gate fails, the selector abstains and the fixed policy is used.

Split conformal prediction guarantees coverage under *exchangeability* [16, 17]. Covariate shift breaks that assumption [18, 19], so the conformal gate is a calibrated instrument in-distribution and a heuristic outside it. It is the support gate, not the conformal gate, that addresses extrapolation, and neither detects concept shift. We state this as a limitation of the design rather than presenting conformal prediction as a solution to out-of-distribution behaviour.

Distribution shift. Eight held-out splits are evaluated separately, never pooled: demand above and below the training range, an interior demand level as an interpolation control, an unseen information lag, an unseen penetration, an unseen green ratio, training without disruption and testing under it, and an incident located on a corridor that carries no disruption in any training context. Pooling these would average an interpolation control together with a domain shift and hide the only thing the experiment can say about them, which is that they differ.

Table 1: The routing-policy portfolio. Every policy is an established family; none is introduced here. The last column states a hypothesis fixed before the experiment.

	Objective	Information used	Intended strength	Hypothesised failure mode
P1	$\min_p L(p)$	none	needs no infrastructure and cannot oscillate	loads the shortest corridor past its capacity
P2	$\min_p \hat{\mu}_p(t - \Delta)$	lagged link travel times	follows congestion as it forms	moves the whole guided cohort together on stale information
P3	$\min_{p \in \mathcal{A}} \max_{e \in p} \rho_e$	lagged travel times and flows, plus its own recent assignments	spreads load when capacity is unevenly distributed	diverts to longer corridors when no capacity shortage exists
P4	$\min_p [\hat{\mu}_p + \lambda \hat{\sigma}_p]$	lagged link travel times	avoids corridors whose travel time is volatile	pays a detour for variance that carries no risk

Table 2: Scenario factors and levels. Demand levels are expressed as a fraction of the measured network capacity, which ranges from 0.31 to 1.10 at the median capacity configuration. Full factorial: 648 contexts.

Factor	Levels	Mechanism it is there to activate
Corridor demand (veh/h)	1200, 1800, 2400, 3000, 3600, 4200	capacity shortage
Green ratio g/C	0.35, 0.50, 0.65	signalised capacity
Guidance penetration	0.2, 0.5, 0.8	controlled cohort size; herding
Information lag Δ (s)	30, 120, 300	staleness; oscillation
Incident regime	none, arterial	travel-time risk
Alternative capacity	1 lane, 2 lanes	whether there is load to balance

7 Results

7.1 Campaign validity

The main campaign executed 12,960 runs over 648 contexts with 4 policies and 5 seeds. No run failed. Every run completed every vehicle it generated (minimum completion rate 1.000), and there were 0 teleports in the entire campaign, so no result rests on a vehicle the simulator removed. All 648 contexts satisfy the validity and completeness rules declared in advance, and none is excluded.

7.2 The portfolio is complementary, and one policy is near-dominant

Gate G1' — declared before the campaign and reported here whichever way it came out — passes: 3 policies (P1, P3, P4) are strictly best in at least one context with a resolved margin. **P2 is never a resolved winner anywhere in the factor space.** Resolved winner counts are P3 227, P4 19 and P1 11.

The winner map is multi-factor but shallow. Always naming the same policy is correct in 64.0% of contexts; the best single factor (demand) reaches 67.0%, and the best pair of factors 71.0%. The regime structure is interpretable and matches the design: P1 wins only where the shortest corridor is uncongested *and* penetration is high enough for the adaptive policies to over-divert; P4 wins at low demand and generous green; P3 wins

policy	objective	information used	intended strength / hypothesised failure mode
P1 Static / shortest	$\min_p L(p)$	none	needs no infrastructure and cannot oscillate loads the shortest corridor past its capacity
P2 Reactive dynamic travel time	$\min_p \hat{\mu}_p(t - \Delta)$	lagged link travel times	follows congestion as it forms moves the whole guided cohort together on stale information
P3 Capacity-aware load balancing	$\min_{p \in \mathcal{A}} \max_{e \in p} \rho_e$	lagged travel times and flows, plus its own recent assignments	spreads load when capacity is unevenly distributed diverts to longer corridors when no capacity shortage exists
P4 Reliability-aware	$\min_p [\hat{\mu}_p + \lambda \hat{\sigma}_p]$	lagged link travel times	avoids corridors whose travel time is volatile pays a detour for variance that carries no risk

$\mathcal{A}$: paths within a declared travel-time tolerance of the best. ρ_e : link utilisation. Δ : information lag. λ : dispersion weight.

Figure 2: The policy portfolio. The two right-hand entries state hypotheses fixed before the experiment; Section 7 reports which were observed.

Table 3: The three decision criteria. All are measured over the same cohort: every vehicle in the network, guided or not, whose intended departure falls in the measurement window.

	Criterion	Measurement
C1	Mean journey time (s/veh)	mean over the cohort of insertion delay plus in-network time. Insertion delay is included: a policy that oversaturates the network produces vehicles that cannot enter it, and excluding their wait would credit the policy for the queue it caused.
C2	CO ₂ emissions (kg)	total CO ₂ over the cohort, HBEFA3 passenger-car petrol Euro-4 class.
C3	Total stopped delay (veh s)	total time spent below 0.1 m/s — the time vehicles spend <i>stationary</i> in queues, as distinct from the time they spend travelling slowly.

every context at the highest demand level.

7.3 Adaptation is worth almost nothing — but the portfolio is worth a great deal

These are different quantities and the difference is the central result.

The single best fixed policy is P3, optimal in 65% of contexts. Running it always costs 474.4 s of system mean journey time per vehicle; the retrospective best-per-context choice costs 473.9 s. The entire value of adapting the policy to the context is therefore **0.45 s per vehicle**, or 0.12%, with a maximum of 17.2 s in any single context.

Choosing the *wrong fixed policy* costs far more. Always running the distance-minimising policy costs +220.2 s per vehicle more than always running the load-balancing policy — +46.4%. The reactive and reliability-aware policies cost +16.3 s (+3.4%) and +18.3 s (+3.9%) respectively.

Getting the fixed policy right is worth up to 46.4% of journey time. Getting the *context-dependent* choice right, given that the fixed policy is already the best one, is worth 0.12%.

7.4 Why: an eighteen-fold asymmetry

The explanation is not that the policies behave alike. It is that the best fixed policy is near-dominant in a specific, measurable sense. In 34.3% of contexts P3 is not the best policy, and when it is not, it loses 1.32 s on

factor	levels	mechanism activated	policy
corridor demand (veh/h)	1200 1800 2400 3000 3600 4200	capacity shortage	P1
green ratio g/C	0.35 0.50 0.65	signalised capacity	P1
guidance penetration	0.2 0.5 0.8	controlled cohort size; herding	P2
information lag (s)	30 120 300	staleness; oscillation	P2
incident regime	none arterial	travel-time risk	P4
alternative capacity	1 lane 2 lanes	whether there is load to balance	P3

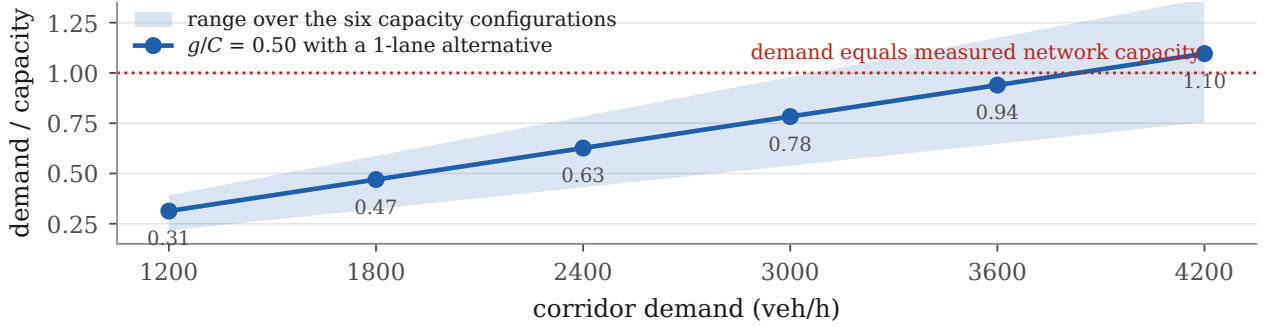


Figure 3: The operating-regime design. Each factor is present to activate a specific policy failure mechanism. Demand levels are set by the *measured* network capacity and span 0.31 to 1.10 of it.

average (at most 17.2 s). In 64.0% of contexts it is the best, and when it is, it wins by 23.1 s on average (at most 153.5 s). Its upside is **18× its downside**.

A portfolio containing such a policy has a large *ranking* structure and a small *decision* structure. Any selector that moves away from it risks a large loss to chase a small gain.

7.5 The effect is at the edge of what the experiment can resolve

Among the 222 contexts where the fixed policy is not best, the mean headroom is 1.32 s against a mean two-standard-error noise band of 1.67 s on the same paired comparison, and the headroom exceeds its own noise in 55 of them (25%). Resolving the campaign-wide mean effect of 0.451 s would require roughly 25 seeds per context — about 64,800 runs. We report this rather than presenting a sub-noise-floor improvement as an achievement.

7.6 The mechanistic rule, and why accuracy is not value

Fitted on all contexts for exposition, the depth-2 mechanistic rule splits first on x_8 , the share of the shortest corridor’s capacity that the guided cohort alone would consume, at a threshold bracketed between 0.748 and 0.781 — reported as a bracket, since the grid cannot place it more finely. The rule is physically readable: when the controlled flow is a small share of the arterial’s capacity the decision turns on the effective green ratio, and when it is a large share it turns on whether the arterial is already oversaturated.

The rule identifies the best policy in 68.8% of contexts, more often than always naming the fixed policy. Its mean decision regret is nevertheless 0.668 s against 0.451 s for the fixed policy. **It is more accurate and more expensive.** Under leave-one-context-out evaluation the same pattern holds: B1 reaches 67% top-1 accuracy against B0’s 65%, and 0.77 s of regret against B0’s 0.45 s.

This is the predict-then-optimize distinction [14] in a traffic setting. A rule trained to name the winner is rewarded for being right in contexts where the margin is negligible and punished only mildly for being wrong in contexts where it is large — exactly the wrong trade under an 18× asymmetry.

7.7 The learned selector works, and it is not worth deploying

Table 5 and Figure 7 give the ladder. Predicting *advantage* rather than identity reverses the failure: B4 reaches 0.20 s of mean regret, closing 55.8% of the gap between the fixed policy and the retrospective best, and it does so while being *less* accurate at naming the winner (73%) than the logistic baseline (76%, 0.28 s). It also has the lowest maximum regret of any selector, 7.0 s against 17.2 s.

The primary comparison declared in advance was B4 against B1. B4 wins it decisively: 0.20 s against 0.77 s. The magnitudes are what matter operationally. B4 improves mean system journey time from 0.45 s of regret to 0.20 s — a saving of 0.252 s per vehicle on a mean journey time of 474.4 s. **That is under a tenth of one per cent.** The selector is real, reproducible, and below the threshold at which any operator would act on it.

7.8 Abstention prevents every out-of-distribution harm, and forgoes the in-distribution gain

The selective selector B5 abstains in 36% of contexts and returns exactly the fixed policy’s regret, 0.45 s. The reason is visible in the mechanism rather than in a tuning choice: the conformal interval on the predicted advantage is wider than the advantage itself, so the requirement that the lower bound exceed zero is never met. **A calibrated gate, correctly applied, refuses to act on an effect smaller than its own uncertainty.**

Table 6 and Figure 8 show why this matters. On the interpolation control (O3) the learned selector helps, cutting regret from 0.56 s to 0.17 s. Under extrapolation and shift it hurts: on unseen high demand it raises regret from 0.00 s to 0.98 s, and under the concept shift of an unseen disruption regime from 0.63 s to 2.22 s. In every one of these cases B5 recovers the fixed policy’s performance exactly.

The support gate behaves as designed: it flags 100%, 100% and 100% of held-out contexts as outside the training envelope on the lag, penetration and green-ratio extrapolations respectively. The mechanistic rule degrades worst of all under shift, reaching 6.15 s on unseen penetration.

7.9 The criteria disagree, and the policies specialise

Over all 648 contexts, 73.0% have a non-singleton Pareto set across the three criteria. Non-domination is not concentrated in one policy: P3 is non-dominated in 86% of contexts, P4 in 81%, P2 in 41% and P1 in 3%.

The criteria are not interchangeable. Journey time and CO₂ correlate at +0.9390 and rank the four policies identically in 52% of contexts; journey time and stopped delay correlate at +0.8916 and agree on the ranking in only 34%. The best policy on CO₂ differs from the best on journey time in 30% of contexts; on stopped delay it differs in 56%.

The policies specialise by criterion rather than by regime. P3 minimises journey time in 418 contexts and P3 minimises CO₂ in 453, but **P4 minimises stopped delay in 429 contexts (66%)** while minimising journey time in only 129. The reliability-aware policy routes towards the steady, unsignalised corridor: its vehicles spend less time stationary and more time moving.

This is where the portfolio’s complementarity actually lives. On the scalar decision cost the portfolio is near-degenerate; on the criterion vector it is not.

Eco-routing remains excluded, on evidence. CO₂ and journey time correlate at +0.9390 and select the same policy in 52% of contexts. An objective nearly monotone in travel time adds a label rather than a policy, and the numbers supporting that judgement are reported here so a reader may disagree with the threshold rather than with an assertion.

7.10 Sensitivity

Across the three declared preference profiles the preferred policy changes in 14.5% of contexts. The journey-time choice is reproduced by the time-weighted profile in 93% of contexts, by the balanced profile in 86% and by the environment-weighted profile in 80%. The decision is therefore stable under reasonable preference assumptions, and where it is not, stopped delay is what moves it.

The two frozen policy parameters behave differently. The dispersion weight of the reliability-aware policy is close to immaterial, changing that policy’s own mean journey time by at most 0.16%; across all four parameter changes the identity of the best policy is unchanged in 78% to 98% of the contexts tested. The admissibility

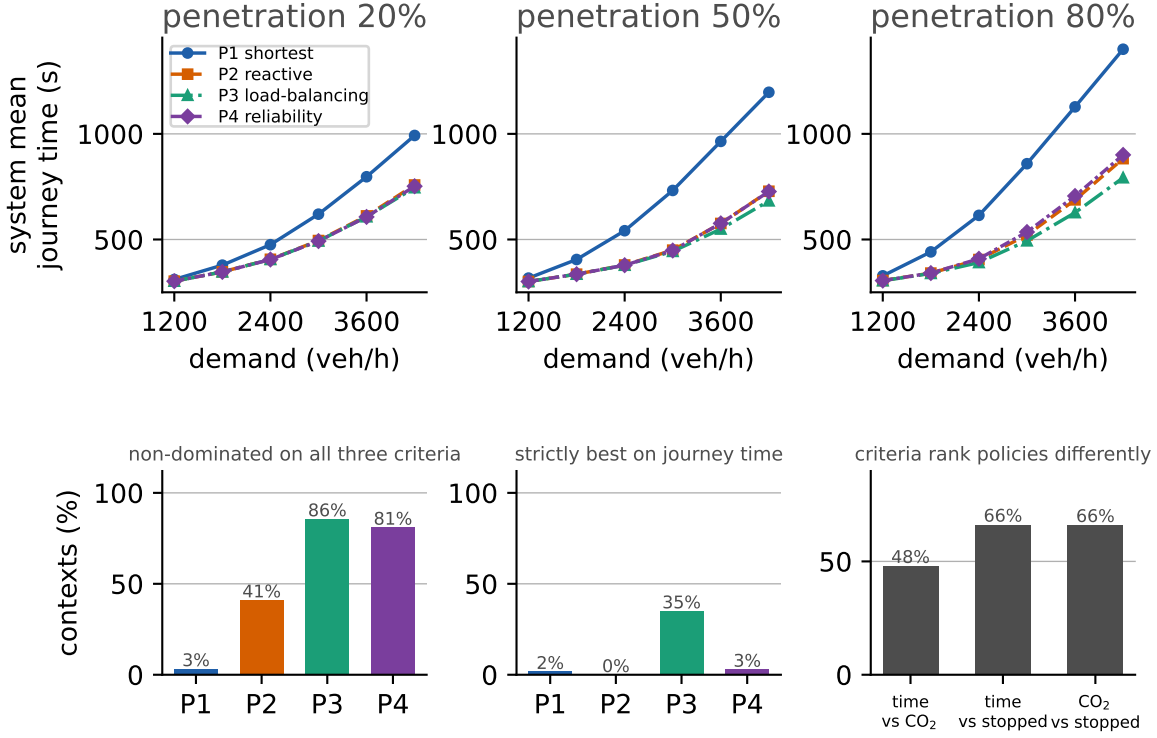


Figure 4: Policy performance and complementarity. Top: system mean journey time against demand at each penetration level. Bottom left: how often each policy is non-dominated across all three criteria. Bottom centre: how often each is strictly best on journey time. Bottom right: how often two criteria rank the policies differently.

tolerance of the load-balancing policy is not: loosening it from 0.20 to 0.30 improves that policy’s own mean journey time by 2.62% (16.7 s). The frozen value therefore *understates* how dominant the load-balancing policy is, which strengthens rather than weakens the central conclusion. We report this rather than retrofitting the better value.

8 Discussion

8.1 What actually drives the choice

The descriptor that organises the winner map is $x_8 = pD/\text{cap}_C$: the share of the shortest corridor’s capacity that the guided cohort alone would consume. It is a product of demand and penetration, and neither alone reproduces the structure — which is why the best single-factor rule reaches only 67.0% against 64.0% for a constant rule. Demand determines whether the corridor is under pressure; penetration determines how much of that pressure a policy commands. A system routing 20% of traffic in a saturated network and one routing 80% in a lightly loaded one are different decision problems, and the descriptor says so.

That is the answer to RQ2: the preference boundary is explicable in mechanistic terms, and the explanation is physical rather than statistical. It is also why the boundary must be quoted as a bracket. The grid samples x_8 at discrete levels, and the transition sits between 0.748 and 0.781. A point estimate would be a property of the fitting procedure, not of the traffic.

8.2 Does machine learning add value over the mechanistic rule?

Yes, and the reason is instructive. The mechanistic rule is *more accurate and more expensive* than doing nothing: it names the best policy more often than the fixed policy does (67% against 65%) and costs more (0.77 s against 0.45 s). The learned selector is *less accurate* than the logistic baseline and cheaper than everything except the retrospective optimum.

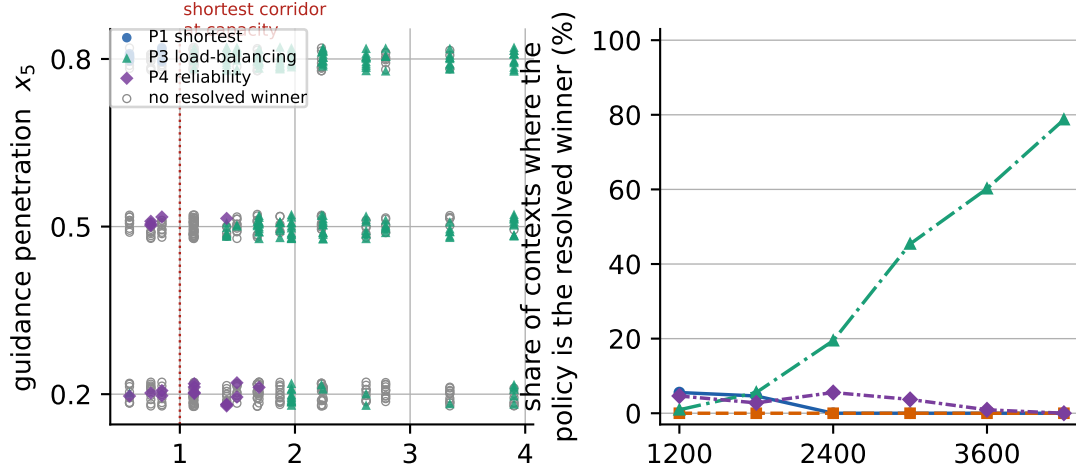


Figure 5: The regime map. Left: the resolved winner in the plane of shortest-corridor saturation against guidance penetration; open circles are contexts where the seeds do not resolve a winner. Right: the share of contexts each policy wins, by demand.

Table 4: Policy complementarity over 648 contexts. A win is *resolved* when the paired seed difference against the runner-up exceeds twice its standard error. Pareto membership is over all three criteria.

	Best on journey time		Non-dominated	
	any margin	resolved	contexts	share
P1	21	11	21	3.2%
P2	80	0	266	41.0%
P3	415	227	555	85.6%
P4	132	19	525	81.0%
<i>Pairwise differences in system mean journey time (s/veh), positive means the first policy is slower</i>				
Pair	mean	5–95% range	outcome-identical	
P1–P2	+203.87	[-0.3, +711.9]	0.0%	
P1–P3	+220.21	[+0.8, +761.6]	0.0%	
P1–P4	+201.87	[-1.0, +711.9]	0.0%	
P2–P3	+16.34	[-0.7, +75.9]	22.7%	
P2–P4	-2.00	[-18.2, +9.6]	42.6%	
P3–P4	-18.34	[-87.3, +1.3]	20.2%	

The difference is not model capacity. It is what the model is asked to predict. B1, B2 and B3 predict the identity of the winner and are trained on a classification loss that treats every context alike. B4 predicts the *advantage* of each policy over the fixed default, so a context where the margin is 20s contributes a hundred times the gradient of one where it is 0.2s. Under an $18\times$ asymmetry between the fixed policy’s upside and its downside, that weighting is the whole game. This is the predict-then-optimize argument [14], and it appears here not as a refinement but as the difference between a selector that helps and one that hurts.

The practical corollary for anyone building such a system: report decision regret, and do not report selection accuracy as though it were a proxy for it. In this study the two rank the methods differently.

8.3 When adaptation is useful, and when a fixed policy is preferable

Adaptation helps in-distribution and hurts outside it. On the interpolation control the learned selector cuts regret from 0.56 s to 0.17 s; on unseen high demand it raises it from 0.00 s to 0.98 s, and under an unseen disruption regime from 0.63 s to 2.22 s. The value of adaptation is a function of regime coverage, not of model capacity, and no amount of capacity substitutes for having seen the regime.

But the decisive observation is one of magnitude. The best deployable selector saves 0.252 s per vehicle against the best fixed policy, on a mean journey time of 474.4 s. Choosing the wrong fixed policy costs 46.4%. The two decisions differ by nearly three orders of magnitude, and only one of them is worth an operator’s

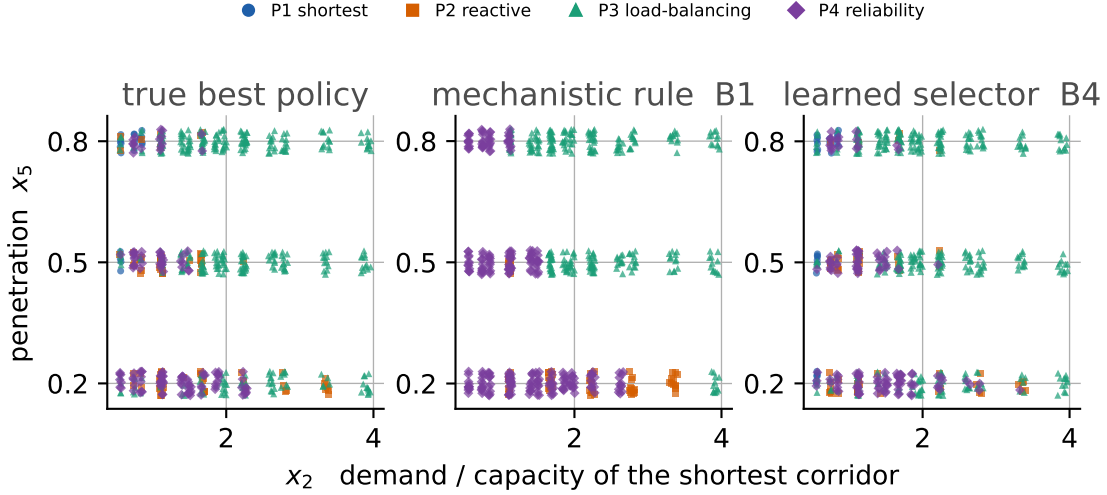


Figure 6: Selection regions in the mechanistic feature space: the true best policy, the depth-2 mechanistic rule, and the learned advantage selector.

Table 5: Decision quality under leave-one-context-out evaluation. Regret is in seconds of system mean journey time per vehicle. “Within noise” is the share of contexts whose regret falls below that context’s own seed noise floor.

		Decision regret (s/veh)				within	gap closed
		mean	median	max	CVaR ₉₀	noise	vs. SBS
B0	best fixed policy (SBS)	0.45	0.00	17.2	3.7	99%	—
B1	mechanistic rule	0.77	0.00	17.2	6.5	96%	-0.714
B2	decision tree, depth 3	0.34	0.00	17.2	3.0	99%	0.238
B3	multinomial logistic	0.28	0.00	17.2	2.5	100%	0.373
B4	GBDT advantage selector	0.20	0.00	7.0	1.7	100%	0.558
B5	selective GBDT with abstention	0.45	0.00	17.2	3.7	99%	0.000
	VBS (retrospective, not deployable)	0.00	0.00	0.0	0.0	100%	1.000

attention.

8.4 What the abstention mechanism is really telling us

B5 abstains in 36% of contexts and never improves on the fixed policy. It would be easy to present this as a failure of the gate, or to loosen it until it acted. It is neither. The conformal interval on the predicted advantage is wider than the advantage, so a gate that requires a positive lower bound cannot certify a gain — and should not. The gate is doing exactly what a calibrated instrument is supposed to do when the effect is smaller than the uncertainty.

The cost of that correctness is the in-distribution gain: B5 forgoes the 0.252 s that B4 achieves. That is the honest trade, and it is the answer to RQ4. A selective mechanism prevented every out-of-distribution harm we constructed, and it did so by declining to act at all.

8.5 Implications for traffic operators

Three, in decreasing order of value.

First, **choose the fixed policy deliberately**. In this environment the gap between the best and worst fixed policy is 46.4% of system journey time. That is where the engineering effort belongs.

Second, **decide which criterion is being optimised, because the policies specialise**. The load-balancing policy minimises journey time in 418 of 648 contexts; the reliability-aware policy minimises stopped delay in 429. An operator whose objective is queue clearance at signalised junctions and one whose objective is corridor journey time should not deploy the same policy, and this is not a preference weighting — it is a different argmin in 56% of contexts.

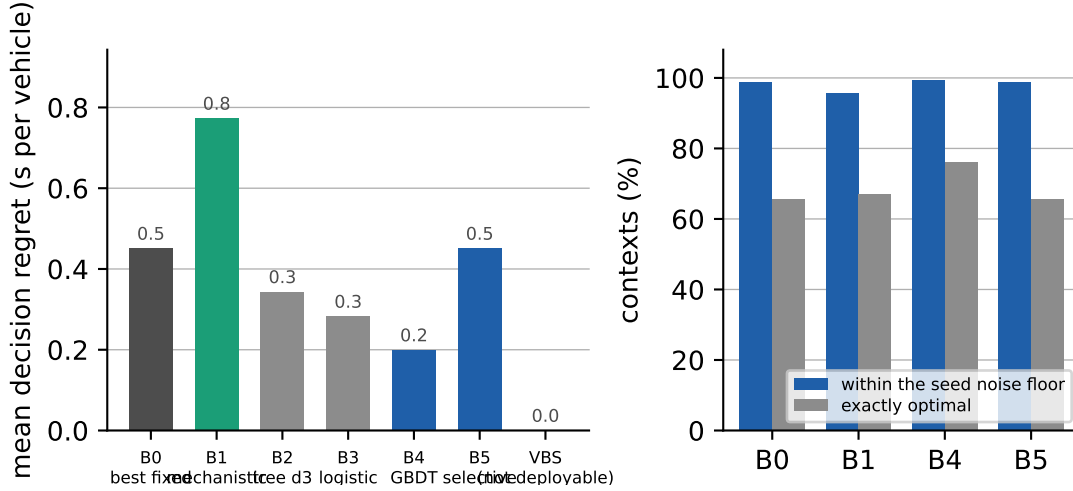


Figure 7: Decision quality. Left: mean regret for each selector, with the retrospective best shown hatched because it is not deployable. Right: the share of contexts in which each selector is within the seed noise floor, and exactly optimal.

Table 6: Distribution shift, reported separately by shift type. Regret in seconds of system mean journey time per vehicle on the held-out regime. The support column is the share of held-out contexts the support gate flagged as outside the training envelope.

	Held out	Shift type	B0	B1	B4	B5	flagged	abstained
O1	demand = 4200.0	extrapolation	0.00	2.35	0.98	0.00	43%	17%
O2	demand = 1200.0	extrapolation	0.53	0.52	0.53	0.53	15%	75%
O3	demand = 2400.0	interpolation	0.56	0.64	0.17	0.56	0%	25%
O4	lag = 300.0	extrapolation	0.23	0.66	0.24	0.23	100%	38%
O5	penetration = 0.8	extrapolation	0.30	6.15	0.29	0.30	100%	14%
O6	gc = 0.65	extrapolation	1.04	3.68	1.16	1.04	100%	51%
O7	incident = 1.0	covariate + concept shift	0.63	5.33	2.22	0.63	32%	36%

Third, **do not build a per-context selector for this decision.** It works, it is reproducible, and it is worth less than a tenth of a per cent. If a selector is built anyway, give it a support gate and an abstention path, because its out-of-distribution behaviour is worse than doing nothing.

8.6 Why this is a useful negative result

The literature on route guidance contains a persistent expectation that context-adaptive selection should pay. What this study shows is that whether it pays is an empirical property of the portfolio, not a general one, and that it can be measured before any selector is built — by comparing the best fixed policy with the retrospective best, and comparing that gap with the noise floor of the same comparison. Both quantities are cheap. Neither requires a model.

Had that comparison been run first here, it would have shown a 0.12% gap sitting below the resolution of a five-seed design, and the selector would not have been built. We suggest it as a standing first step: *measure the headroom and the noise before choosing a model class.*

9 Limitations

The environment is synthetic and the study claims no external validity. The network is a controlled instrument built to isolate mechanisms, not a model of any city. Every result is scoped to it. What is portable is the method rather than the numbers: the descriptors are dimensionless, so a boundary is stated as a degree of saturation and a controlled-flow share rather than as a count of vehicles per hour, and both are measurable on a real network. No claim of real-world validation is made anywhere in this paper.

The context is assumed observable. The selector is given the operating conditions. A deployed system would have to forecast them, and would inherit the forecast’s error on top of the selector’s. Forecasting the

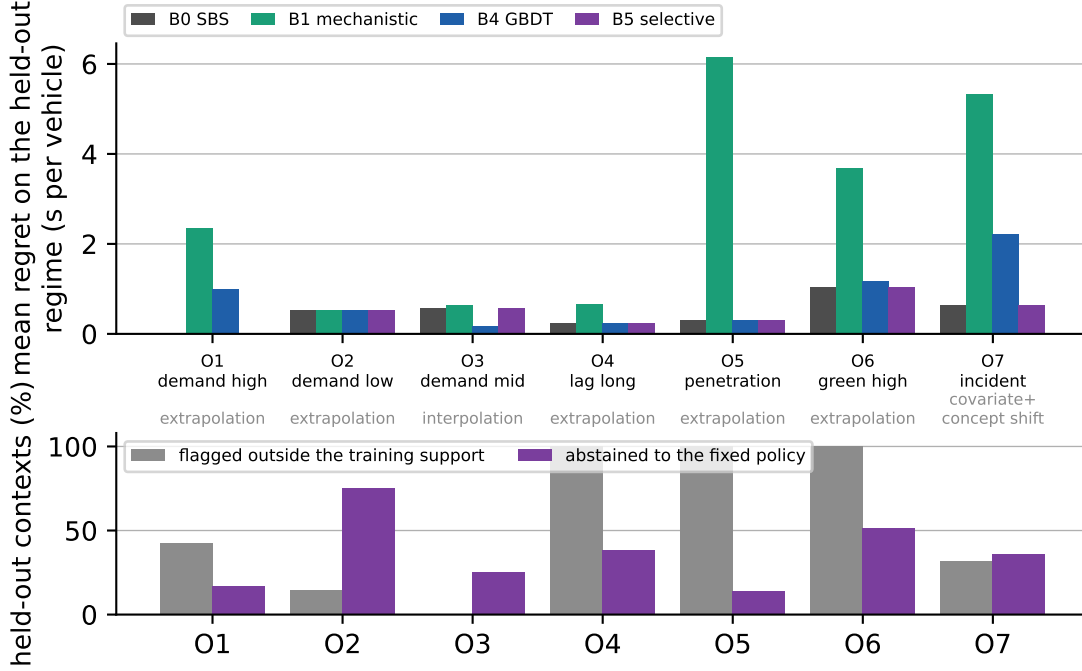


Figure 8: Distribution shift, reported separately by shift type. Top: mean regret on the held-out regime. Bottom: the share of held-out contexts flagged outside the training support, and the share on which the selective selector abstained.

context is a separate problem and is not addressed here. The reported decision quality is therefore an upper bound on what the same selector would achieve behind a forecast.

One routed origin–destination pair. Corridor demand is a single OD pair choosing among three corridors. The four minor streets carry independent demand through the same signals, which is what couples the routing decision to traffic that is not being routed, but the decision itself is a three-way choice on one OD pair. Many simultaneously-routed OD pairs, with the interactions between their choices, is the clearest extension and is not covered by anything reported here.

The uncertainty guarantee is conditional and partial. Split conformal coverage requires exchangeability, which covariate shift breaks [18, 19]. The conformal gate is a calibrated instrument in-distribution and a heuristic outside it; the support gate is what addresses extrapolation; and neither detects concept shift — a context can sit comfortably inside the training support while the relationship between context and outcome has changed. We report coverage only where its assumption holds, and report gate behaviour rather than guarantees elsewhere.

Grid resolution bounds every boundary claim. The experiment samples a grid. A boundary is reported as a bracket between adjacent sampled levels and never as a continuous threshold, even after refinement, because a grid cannot locate a boundary more finely than its own spacing.

No causal identification. Where a factor is associated with a change of preferred policy, that is an association within a designed factorial, not an identified causal mechanism. The factorial licenses the statement that the preference differs across levels of a factor with the others held fixed; it does not license a claim about why, beyond the mechanistic reasoning that motivated the design, which is a hypothesis the experiment is consistent with rather than a result it establishes.

Policy parameters are fixed points, not optima. The admissibility tolerance of the load-balancing policy and the dispersion weight of the reliability-aware policy were frozen before the experiment and varied afterwards only for sensitivity. Neither was optimised, and a better-tuned version of either could change its standing. We report the sensitivity rather than claiming the frozen values are the best available.

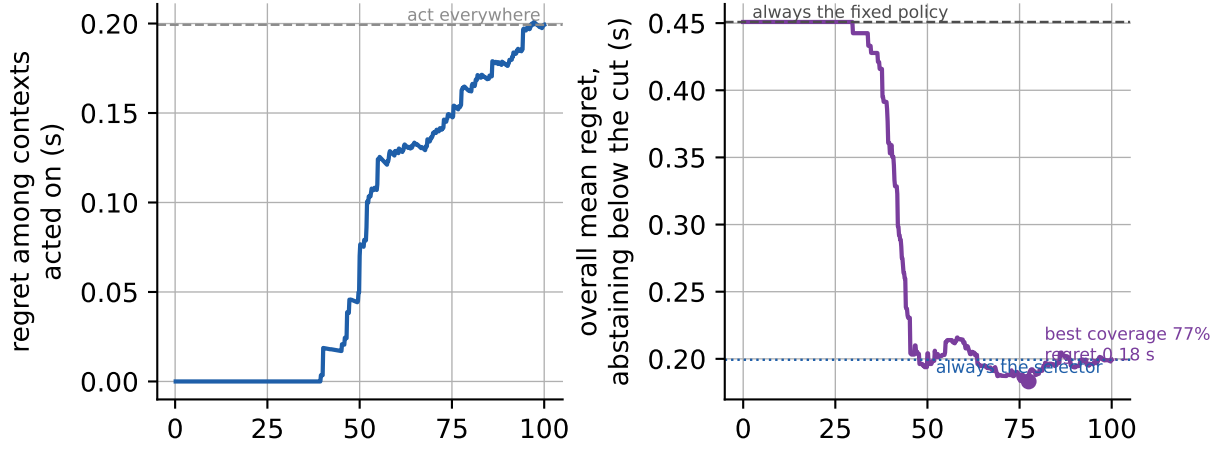


Figure 9: Risk-coverage behaviour. Left: mean regret among the contexts acted on, as coverage increases from the most confident. Right: mean regret over all contexts when the selector abstains below the coverage cut.

The emission model is a simulator component. CO₂ is computed by the simulator’s HBEFA3 implementation for one vehicle class [22]. Fleet composition, cold starts and gradient are not modelled, so C2 should be read as a consistent comparative index across policies in this environment rather than as an absolute emissions estimate.

No monetary valuation. No verified value of time, value of reliability or carbon price was available for this study, so none is used. The criteria are compared by Pareto dominance, and scalar sensitivity uses declared normalised weight profiles. This keeps the study free of invented constants, but it also means the paper cannot express a trade-off in currency, which is what an operator would ultimately want.

10 Conclusion

We characterised context-dependent selection among four established urban routing policies in 12,960 microsimulation runs over 648 controlled contexts, with every policy decision made from information available before activation and every comparison paired under common random numbers.

The portfolio is genuinely complementary: three of the four policies are strictly best in some context by a margin the seeds resolve, the winner map depends on more than one factor, and 73.0% of contexts have a non-singleton Pareto set across journey time, CO₂ and stopped delay. The reliability-aware policy, which almost never minimises journey time, minimises stopped delay in 66% of contexts. Complementarity in this portfolio lives in the criterion vector, not in the scalar cost.

The value of exploiting it is nevertheless 0.45 s of system mean journey time per vehicle — 0.12% — because the best fixed policy is near-dominant, with an upside 18× its downside. Against this, choosing the wrong fixed policy costs 46.4%.

Within that small margin the methodological findings are clear. A selector trained to predict the winner’s identity is more accurate than the fixed policy and more expensive than it; a selector trained to predict each policy’s advantage over the fixed default is less accurate and cheaper, closing 55.8% of the available gap. Adaptation helps in-distribution and harms under every extrapolation and shift we constructed, and a support-and- confidence gate prevented all of that harm by declining to act — correctly, because the uncertainty exceeds the effect.

The operational conclusion is not that adaptive routing-policy selection cannot work. It is that in this environment the decision worth making is which fixed policy to run and which criterion to optimise, and that the headroom for per-context adaptation can be measured, cheaply and before any model is built, by comparing the best fixed policy against the retrospective best and both against the noise floor of the same comparison.

Reproducibility

All scenario, policy, simulation and analysis code, every run record, and the scripts that generate every number, table and figure in this paper are in the accompanying repository. The experiment specification was committed before any evaluation run was executed, and one protocol deviation is recorded with its reasons.

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